Potential Flow and Bernoulli's Theorem

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Constant density Euler Equations

Momentum $$\frac{D\vec{u}}{Dt} = -\frac{1}{\rho_0}\nabla p - g\hat{k}$$

Mass $$\vec{\nabla}\cdot\vec{u} = 0$$

Dropping viscosity lowers the order of spatial derivatives by one. Analytical solutions are hard to find, so the point of the exercise is to show off a general problem reduction procedure.

We also introduce a fair amount of machinery to describe vector fields along the way.

Constant density Euler Equations

Momentum $$\frac{D\vec{u}}{Dt} = -\frac{1}{\rho_0}\nabla p - g\hat{k}$$

Mass $$\vec{\nabla}\cdot\vec{u} = 0$$

The game from viscous fluids of balancing different types of terms is difficult to play with the Euler equations because we invariably have to deal with he nonlinear terms. A variety of vector calculus tools have been used to reduced flows to something tractable instead.

The Stream Function

In 2D, say (x,z), the most important equation turns out to be the conservation of mass.

$$\boxed{\vec{\nabla} \cdot \vec{u} = 0}$$

$$\vec{u} = (u(x, z, t), w(x, z, t))$$

Vector calculus then guarantees the existence of a function called the stream function, from which the velocity components may be calculated as:

$$(u(x, z, t), w(x, z, t)) = \left(\frac{\partial \psi}{\partial z}, -\frac{\partial \psi}{\partial x} \right)$$

This also means that the stream lines are the level curves of psi or the tangent curves of the velocity field.

$$\psi(x, z, t) = C$$

The Potential

If we think of the conservation of mass as a restriction on the vector field (i.e. the divergence free condition) we can also use the cross-product to restrict the vector field.

$$\boxed{\vec{\nabla} \times \vec{u} = \vec{0}}$$

Vector calculus then guarantees the existence of a function called the potential, from which the velocity components may be calculated as:

$$(u(\vec{x}, t), v(\vec{x}, t), w(\vec{x}, t)) = \left(\frac{\partial \phi}{\partial x}, \frac{\partial \phi}{\partial y}, \frac{\partial \phi}{\partial z} \right)$$

This means that for certain classes of vector fields the governing equations could be simplified without any concern for the actual individual physical situation.

Absorbing pressure: the one simple balance we can form in the Euler equations

$$0 = -\frac{\partial p_H}{\partial z} - \rho_0 g$$

hydrostatic balance

$$Let, p = p_H + p_D$$

$$\rho_0 \frac{D\vec{u}}{Dt} = -\nabla p_H - \nabla p_D - \rho_0 g \hat{k}$$

these balance

$$\boxed{\rho_0 \frac{D\vec{u}}{Dt}} = \boxed{-\nabla p_D}$$

And we usually drop the subscript D

$$\boxed{\vec{\nabla} \cdot \vec{u} = 0}$$

Conservation of mass, just for completeness

$$\vec{\nabla} \cdot \vec{u} = 0$$

$$\vec{\nabla} \times \vec{u} = \vec{0}$$

If we assume that the flow is irrotational, we can derive a much easier set of equations to deal with

$$\vec{\nabla} \times \vec{u} = \vec{0} \implies \vec{u} = \nabla \phi$$

From vector calculus we get that the velocity is derived from a potential. And then from conservation of mass we get:

$$\vec{\nabla} \cdot \vec{u} = \vec{\nabla} \cdot \vec{\nabla} \phi = \nabla^2 \phi = 0$$

So irrotational, incompressible flow is governed by the linear Laplace equation!
This observation dominated fluid mechanics for a century with neat links to the theory of complex variables:
https://en.wikipedia.org/wiki/Potential_flow

- In the complex variables version of the theory, the potential and stream function are the real and imaginary parts of a complex valued potential.
- The linearity of the Laplace equation means that simple functions can be added together to form more complex flows.
- This can be a lot of fun to play with.
- The question of how to recover a physical variable like the pressure is a bit trickier and needs Bernoulli's theorem.

https://potentialflow.com/

Practical Questions

- Is a velocity field one for potential flow? If yes then it must be both incompressible and irrotational.
- Since potential flow involves solutions of Laplace's equation, what are the exceptional solutions that are worth knowing?
- Examples include: sources and sinks, the irrotational vortex and the doublet.

Constant density Euler Equations

Momentum $$\frac{D\vec{u}}{Dt} = -\frac{1}{\rho_0}\nabla p - g\hat{k}$$

Mass $$\vec{\nabla}\cdot\vec{u} = 0$$

We have discussed potential flow, or flow of inviscid, incompressible, irrotational fluids.
The problem of finding the pressure was left hanging.
This is important because pressure is relatively easy to measure.
Here we develop a classical "first integral" namely Bernoulli's theorem.

Bernoulli's equation: a first integral of the Euler equations

$$\vec{u} \cdot \nabla \vec{u} = -(\vec{u} \times \vec{\omega}) + \frac{1}{2}\nabla(q^2)$$

vector identity for nonlinear terms

$$q^2 = \vec{u} \cdot \vec{u}$$

$$B = \frac{1}{2}q^2 + \int \frac{dp}{\rho} + gz$$

$$\vec{u}_t + \nabla B = \vec{u} \times \vec{\omega}$$

Rewritten Euler equations

- Vector or tensor identities allow us to rewrite the nonlinear terms in the Navier-Stokes equations.
- We then define the Bernoulli function B.
- Finally we use certain simplifying assumptions to get an equation of the form "gradient of something equals zero".

Bernoulli: Steady flow

$$\nabla B = (\vec{u} \times \vec{\omega})$$

Rewritten steady Euler equations

$$\frac{1}{2}q^2 + \int \frac{dp}{\rho} + gz \text{ is constant along a streamline}$$

- When the flow is steady the equation simplifies.
- The left hand side is a vector perpendicular to lines of constant B.
- The right hand side is a vector perpendicular to both $\vec{u}$ and the vorticity, $\vec{\omega}$, or in other words perpendicular to streamlines and vortex lines.
- Thus the Bernoulli function is constant along a streamline.
- This is useful, but for example when one derives the theory of surface waves it is not enough.

Bernoulli: Steady irrotational flow

$$\nabla B = 0$$

Rewritten steady Euler equations with vorticity zero

$$B = C_1$$

Constant density is the simplest case so let's consider that

$$\frac{1}{2}\rho_0 \vec{u} \cdot \vec{u} + p + \rho_0 g z = C_1$$

Could simplify using the hydrostatic pressure

$$\frac{1}{2}\rho_0 \vec{u} \cdot \vec{u} + p_d + p_H(z) + \rho_0 g z = C_1$$

$$\frac{dp_H}{dz} = -\rho_0 g z$$

$$\frac{1}{2}\rho_0 \vec{u} \cdot \vec{u} + p_d = C_1$$

So higher velocity implies a drop in pressure!

Exercise: Look up demos of Bernoulli effects on YouTube and recreate one for yourself.

Bernoulli: Unsteady Potential Flow

$$\vec{u} = \nabla \phi$$

$$\nabla \left(\phi_t + B \right) = 0$$

$$\phi_t + \frac{1}{2} q^2 + \int \frac{dp}{\rho} + gz = F(t)$$

because potential flow is irrotational

- When the flow is irrotational the equation simplifies again, albeit differently.
- If it is also incompressible we find that the velocity can be derived from a potential (hence potential flow).
- Now when density is constant, a different algebraic equation results one that can sometimes take into account time dependence (this is used in surface water wave theory, for example).

$$\rho_0 \phi_t + \frac{1}{2} \rho_0 \vec{u} \cdot \vec{u} + p + \rho_0 g z = F(t)$$

$$\boxed{\phi_t + \frac{1}{2} |\nabla \phi|^2 + \frac{p}{\rho_0} + gz = F(t)}$$

$F(t)$ is arbitrary so we absorb the factor of ρ_0

Practical Concerns

- Bernoulli's equation is a staple of pre-calculus, or formula based physics.
- It can be useful in simple situations, and in capable hands can crack open even complex problems.
- Theoretically it is an example of "first integrals" of PDEs, which is a powerful tool.
- It can be difficult to know in practice when the conditions needed for Bernoulli's theorem are met in real world situations.